User Related Command

- Creates a new user named username.
- It also creates a home directory (/home/username) and prompts for a password.
    - Syntax:
        - $sudo adduser username
        - e.g $sudo adduser john
            - Creates a new user named john.
            - Prompts to set a password and fill in additional information (which can be skipped).
            - Creates a home directory at /home/john.
            - Sets appropriate permissions for the home directory.
            - Adds the user to default groups (like john's primary group).
- To add a user to a specific group
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        - Syntax:
        - $sudo adduser username groupname
        - e.g $sudo adduser john grp1
            - Adds the user john to the group grp1.
            - Adds the user john to the sudo group, granting administrative privileges.
- To set or change a user's password
    - $sudo passwd username

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1. List All User
   $cat /etc/passwd
2. Show Only Group Names
   $cut -d: -f1 /etc/passwd
3. List Currently Logged-in Users
   $who or $whoami

- To delete a user from the system
    - $sudo deluser username
        - Removes username from the system, but their home directory and files remain.
    - $sudo deluser --remove-home username
        - Deletes username and removes their home directory.

Group Related Command

Checking Group Membership

- Creating Groups
    - syntax:
    - $sudo addgroup groupname
    - e.g $sudo addgroup grp1
        - Creates a new group called groupname.
- Deleting Groups
    - syntax:
        - $sudo delgroup groupname
            - Deletes the group groupname (only if no users are in it).
            - e.g $sudo delgroup grp1
                - Deletes the group grp1.
- Displays the groups of the current user.
    - $groups
    - e.g $groups
- Displays the groups of a specific user.
    - $groups username
    - e.g $groups john
        - Displays the groups that the user john belongs to.

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1. List All Groups
   $cat /etc/group
2. Show Only Group Names
   $cut -d: -f1 /etc/group
3. List Groups for a Specific User
   $groups username

- Modifying User Group Membership
    - $sudo usermod -a -G groupname username
        - Adds username to the secondary group groupname.
        - The -a (append) option is important; without it, the user would be removed from other groups.
    - $sudo usermod -g groupname username
        - Changes the primary group of username to groupname.

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chgrp

- It is used to change the group ownership of a file or directory.
- You must be the file owner or have sudo privileges to change the group.

- The specified group must already exist on the system.
- Syntax:
    - For changing the group of a file
        - $chgrp group file
        - e.g $chgrp grp1 a.txt
          -Changes the group ownership of a.txt to grp1.
    - For changing the group of a directory
        - $chgrp -R groupname directory
            - (-R for Recursive)
            - This changes the group of the directory and all files/subdirectories inside it.

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su

- Switch User
    - The su (substitute user) command allows you to switch to another user account in the terminal.
    - By default, it switches to the root user if no username is provided.
- Switch to another user (requires target user's password)
    - Syntax:
        - $su username
        - e.g $su john
            - Switches to the user john and prompts for john's password.

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chown

- It is used to change the owner or group of a file or directory.
    1. Change Only File Owner
       $sudo chown user file
       -Changes the owner of file to user and group remains unchanged.
    2. Change Only the Group
       $sudo chown :group file
       -Changes the group of file to group without modifying the owner.
    3. Change Both Owner and Group
       $sudo chown user:group file
       -Changes the owner to user and the group to group.
    4. Change Ownership Recursively (-R)
       $sudo chown -R user file
       -Changes the owner of file and all its contents (if it's a directory).
    5. Change Ownership of a Symbolic Link (-h)
       $sudo chown -h user file
       -Changes the owner of a symbolic link, not the file it points to.